

ITR0 Document

Table of Contents

ITR0 Document 1

Vision Statement2

Big User Stories3

Detailed User Stories for Itr15

Story Mapping.....7

Customer meeting’s video summary7

Vision Statement

StoreFlow Analytics Vision Statement

The StoreFlow Analytics System will be used by independent grocery store owners and managers to measure in-store customer traffic, movement, and purchasing outcomes so they can understand conversion rates, evaluate product advertising, and improve store operations. The system combines overhead camera-based path tracking with point-of-sale (POS) sales data to produce actionable reports while operating locally and using anonymized, aggregate data.

The system is designed for brick-and-mortar grocery retailers who want to answer practical questions such as: How many people enter the store? How many end up purchasing? Where do customers spend time? Do endcaps or shelf promotions affect purchasing? Where do bottlenecks form (aisles, sampling stands, checkout lines)?

StoreFlow Analytics will provide three core capabilities. First, it will support a virtual 2D store layout that represents the physical store, allowing staff to label key zones such as aisles, shelves, promotional displays, sampling stands, and checkout areas.

Second, the system will perform in-store tracking to estimate customer counts and movement paths through the store, including where customers stop, dwell, or queue. The system will also integrate these movement signals with sales outcomes by linking storewide traffic patterns with POS data collected at checkout

Third, the system will transform collected signals into actionable business insights through reports and summaries. These insights will include conversion-focused measures (traffic vs. purchases), promotion effectiveness (exposure vs. purchase), operational metrics (checkout wait times, peak periods), and layout/flow indicators (crowding and bottlenecks).

The project will be considered successful based on the following criteria. First, the store owner can use the system to obtain a clear conversion estimate (visitors vs. purchasers) over selected time periods. Second, the system can produce interpretable reports that allow the owner to make at least one concrete decision such as adjusting staffing coverage for peak checkout times or modifying a product advertisement placement. Third, the system demonstrates practical deployability for an independent grocer by running locally, leveraging store infrastructure, and presenting results in a way the owner finds useful and credible. If these criteria are met and the customer indicates they would prefer using this approach over their current guesswork process for layout, promotions, and scheduling, the system will be considered a success.

Big User Stories

Store Layout/Mapping

As a store owner, I want to have a top-down digital 2D layout of my store with detailed labelled product zones with customer movement accurately mapped onto it.

Priority: High

Cost: 12 Days

Analytics and Actionable Insights

As a store owner, I want processed analytics on cashier wait wait times, foot-traffic congestion, and product interaction so that I can improve store layout.

Priority: High

Cost: 10 Days

Sales and Behavior Data Integration

As a store owner, I want to combine customer movement data with cashier sales data so that I can understand the conversion rate between what customers look at and what they actually buy.

Priority: High

Cost: 14 Days

Customer Movement/Path Tracking

As a store owner, I want to monitor how customers move throughout my store so that I can understand traffic patterns and travel behavior.

Priority: High

Cost: 4 Days

Detailed User Stories for Itr1

Compiled Binaries for Windows (X86)

.exe file for user to execute and test program. User can operate software on his own.

Priority: High

Cost: 1.5 Days

Mock Sales Database

Mock cashier sales database for storing, retrieving, and analysing sales data

Priority: High

Cost: 4 Days

Wiki/ReadMe

Help wiki (ReadMe) explaining system features.

Priority: Medium

Cost: 1.5 Days

Basic Camera Feed Display

Display live camera feed within interface to observe store activity in real time.

Priority: Medium

Cost: 2 Days

Basic GUI Setup

Graphical user interface that displays a live camera feed, input controls, system logs, theme colour selector, and sales data CSV import button.

Priority: High

Cost: 3 Days

Story Mapping

- Big Stories
 - Store Layout/Mapping
 - IT1
 - Analytics and Actionable Insights
 - IT2
 - Sales and Behavior Data Integration
 - ITR3
 - Customer Movement/Path Tracking
 - IT4
- Detailed User Stories (ITR1)
 - Compiled Binaries for Windows (X86)
 - Ryan
 - Mock Sales Database
 - Mani
 - Wiki/ReadMe
 - Jeduson
 - Basic Camera Feed Display
 - Ryan
 - Basic GUI Setup
 - Ryan

Customer meeting's video summary

< <https://youtu.be/F5pui8nAKh4> >